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Time taken 10 mins 15 secs

Grade 10.00 out of 10.00 (100%)

Question 1

Correct

Mark 1.00 out of 1.00

Ordinal Encoding should be used when categorical variables have a natural order.

- ☒ True ✓
- ☐ False

The correct answer is 'True'.

Question 2

Correct

Mark 1.00 out of 1.00

Outliers differ from noise because:

- ☐ a. None of them
- ☐ b. Outliers must always be kept
- ☒ c. Outliers are extreme observations far from others ✓
- ☐ d. Noise is systematic, outliers are random

Your answer is correct.

The correct answer is: Outliers are extreme observations far from others

Question 3

Correct

Mark 1.00 out of 1.00

Ridge regression can completely eliminate irrelevant features by setting coefficients to zero.

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question 4

Correct

Mark 1.00 out of 1.00

Which imputation method replaces missing values using nearby data points?

- ☒ a. KNN imputation ✓
- ☐ b. median imputation
- ☐ c. Deletion
- ☐ d. Mean imputation

Your answer is correct.

The correct answer is: KNN imputation

Question 5

Correct

Mark 1.00 out of 1.00

Why is StandardScaler(with_mean=False) used with PolynomialFeatures?

- ☐ a. To speed up convergence only
- ☒ b. To preserve sparsity of feature matrix ✓
- ☐ c. To remove outliers
- ☐ d. To improve accuracy

Your answer is correct.

The correct answer is: To preserve sparsity of feature matrix

Question 6

Correct

Mark 1.00 out of 1.00

Increasing the size of the training dataset always reduces overfitting.

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question 7

Correct

Mark 1.00 out of 1.00

Which loss function is commonly used in linear regression?

- ☒ a. Least squares loss ✓
- ☐ b. None of them
- ☐ c. Log loss
- ☐ d. Cross-entropy loss

Your answer is correct.

The correct answer is: Least squares loss

Question 8

Correct

Mark 1.00 out of 1.00

Exploratory Data Analysis (EDA) is mainly used to tune model hyperparameters.

- ☐ True
- ☒ False ✓

The correct answer is 'False'.

Question 9

Correct

Mark 1.00 out of 1.00

A model has a training error of 20 and a test error of 22, then it suffers from _____

- ☐ a. High variance
- ☒ b. High bias ✓
- ☐ c. Low bias
- ☐ d. Low variance

Your answer is correct.

The correct answer is: High bias

Question 10

Correct

Mark 1.00 out of 1.00

Regularization helps reduce overfitting by penalizing large model coefficients.

- ☒ True ✓
- ☐ False

The correct answer is 'True'.